

Answer the questions in the spaces provided. If you run out of room for an answer, continue on the back of the page.

Name and section: _____

Instructor's name: _____

1. Is it true that $x^n + y^n = z^n$ if x, y, z and n are positive integers?. Explain.
2. Prove that the real part of all non-trivial zeros of the function $\zeta(z)$ is $\frac{1}{2}$
3. Compute

$$\int_0^{\infty} \frac{\sin(x)}{x}$$